

Week 1: Data Understanding & Analysis Report

AI-Powered Data Analysis Remote Internship — Excelerate

RIT Opportunity Wise — Combined Team Submission

Source: RIT_cleaned.csv (8,558 records, Jan 2023 – Mar 2024)

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1. Dataset Overview

The dataset used in this project tracks student engagement and participation across various learning opportunities offered through the RIT platform. It captures learner sign-ups, applications, and progress across 23 unique opportunities spanning five categories: Internship, Course, Event, Competition, and Engagement.

Each row represents a single learner's interaction with one opportunity, including when they signed up, applied, and their current participation status (e.g. Started, Rejected, Dropped Out). The dataset also captures demographic information such as gender, country, institution, and field of study, which can help explain patterns in participation across different learner groups.

- Rows: 8,558
- Number of variables (columns): 16
- Unique opportunities: 23
- Date range covered: 5 October 2022 – 11 March 2024 (Apply Date)

2. Data Quality Assessment and Cleaning

This section summarizes the assessment and cleaning of the RIT Opportunity Wise dataset, performed in R. The workflow evaluated missing values, duplicate records, invalid dates, formatting inconsistencies, categorical variables, and age outliers before applying appropriate cleaning techniques.

Data Quality Assessment Summary

Issue	Before	Action	After
Missing Values	4 columns with missing values	Categorical values replaced with placeholders; genuine date NAs retained	5,389 values handled
Duplicate Rows	0 duplicates	Verified using duplicated() and distinct()	0 duplicates
Extra Spaces	Present	trimws() / str_trim()	None
Gender Formatting	Inconsistent	Standardized	Consistent
Country Formatting	Inconsistent	Standardized	Consistent
Invalid Dates	5,262 records	Corrected / Flagged	5,262 corrected
Name Cleanup	Special characters / spaces	Trimmed	Cleaned
Status Codes	All valid	Validated	Consistent
Date Fields	Multiple formats	Converted to POSIXct / Date	Standardized
Outliers (Age)	1,007 flagged	Flagged for review	Retained

Missing Values

Missing values were assessed using colSums(is.na()). Missing Gender values were replaced

with 'Not Specified', Institution Name with 'Unknown', and Current/Intended Major with 'Not Specified'. Genuine missing dates were preserved as NA to maintain data integrity.

Duplicate Records

Duplicate rows were checked using duplicated() and verified with distinct(). No fully duplicated records remained after assessment.

Date Columns

Apply Date, Date of Birth, Opportunity Start Date, Opportunity End Date, and Learner SignUp DateTime were converted into standard Date/POSIXct formats. Invalid formats were corrected where possible and flagged otherwise.

Categorical Standardization

Gender values were standardized to Male, Female, and Not Specified. Country names were standardized, and extra spaces were removed from text fields including First Name, Country, Opportunity Category, and Status Description.

Status Codes

Status codes were compared with predefined valid codes. No invalid codes required correction.

Age Outliers

Age was derived from Date of Birth and assessed using the IQR method. Extreme values were flagged for review rather than deleted.

Summary of Work Completed

- Profiled missing values and calculated percentages.
- Handled missing categorical values using placeholders.
- Verified duplicate records.
- Standardized date formats.
- Standardized gender and country values.

- Removed extra spaces and cleaned text fields.
- Validated status codes.
- Identified age outliers using the IQR method.

The assessment improved the dataset's completeness, consistency, and reliability while preserving valid information. The cleaned dataset is suitable for analysis and reporting.

3. Data Dictionary

The table below documents each variable in the dataset, its meaning, data type, and analytical relevance.

#	Variable Name	Description	Data Type	Analytical Relevance
1	Learner SignUp DateTime	Date and time the learner registered on the platform	Date/Time	Analyzes registration trends and onboarding growth over time
2	Opportunity ID	Unique identifier assigned to each opportunity	Text (UUID)	Uniquely identifies opportunities and links records across the dataset
3	Opportunity Name	Name/title of the opportunity the learner applied for	Text (String)	Identifies popular opportunities and supports application trend analysis
4	Opportunity Category	Classification of the opportunity (Internship, Course, Event, Competition, Engagement)	Categorical (Text)	Enables comparison of participation and outcomes across opportunity types
5	Opportunity Start Date	Date and time the opportunity begins	Date/Time	Used to analyze opportunity timelines and scheduling patterns

6	Opportunity End Date	Scheduled end date of the opportunity	Date/Time	Used to analyze opportunity duration and completion windows
7	First Name	First name of the learner	Text (String)	Primarily for identification; generally excluded from analytical models for privacy
8	Date of Birth	Learner's date of birth	Date/Time	Can be used to derive age for demographic analysis
9	Gender	Learner's gender	Categorical (Text)	Supports demographic breakdown of participation and outcomes
10	Country	Learner's country of residence	Categorical (Text)	Enables geographic analysis of learner reach and engagement
11	Institution Name	Name of the learner's educational institution	Text (String)	Supports institutional-level participation analysis
12	Current/Intended Major	Learner's current or intended field of study	Text (String)	Helps identify which fields of study engage most with each opportunity type
13	Apply Date	Date and time the learner submitted the application	Date/Time	Used to analyze application timing and volume trends
14	Entry Created At	Date and time the application record was created	Date/Time	Supports data-quality checks and record audit

				trails
15	Status Description	Description of the learner's application status (e.g. Started, Rejected, Dropped Out)	Categorical (Text)	Core field for tracking participation outcomes and funnel drop-off
16	Status Code	Numeric code representing the application status	Text (String)	Machine-readable counterpart to Status Description; stored as text since no math is performed on it

4. Historical Analysis

Overall Application Trend

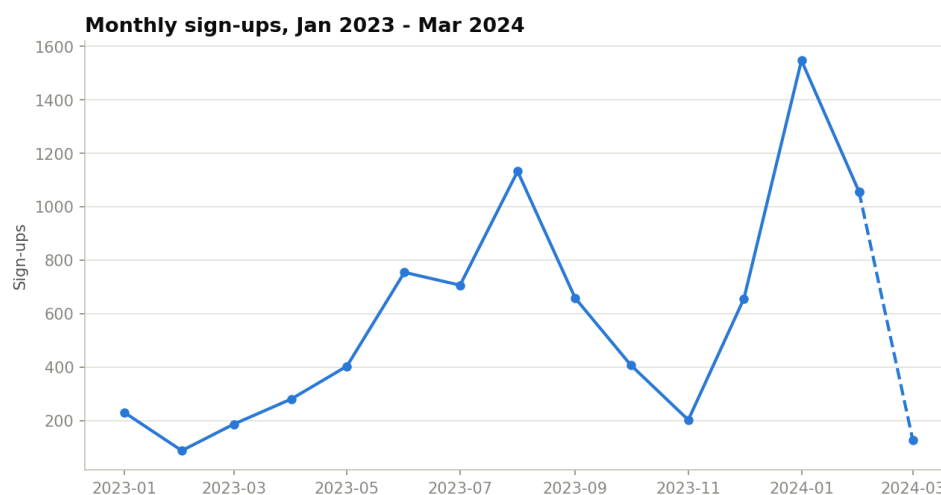


Figure 3 — Monthly sign-ups, Jan 2023 – Mar 2024

Growth / Decline Pattern

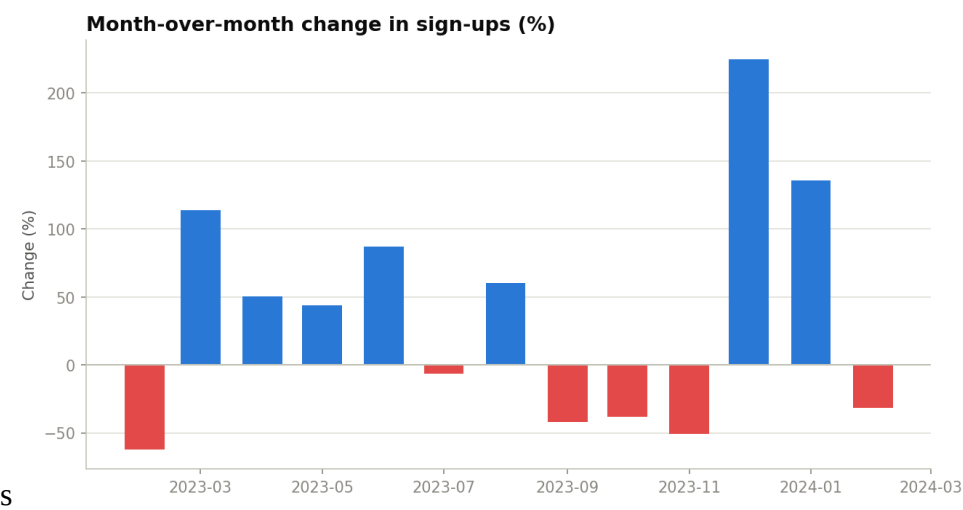


Figure 4 — Month-over-month change in sign-ups (%)

Trend by Category

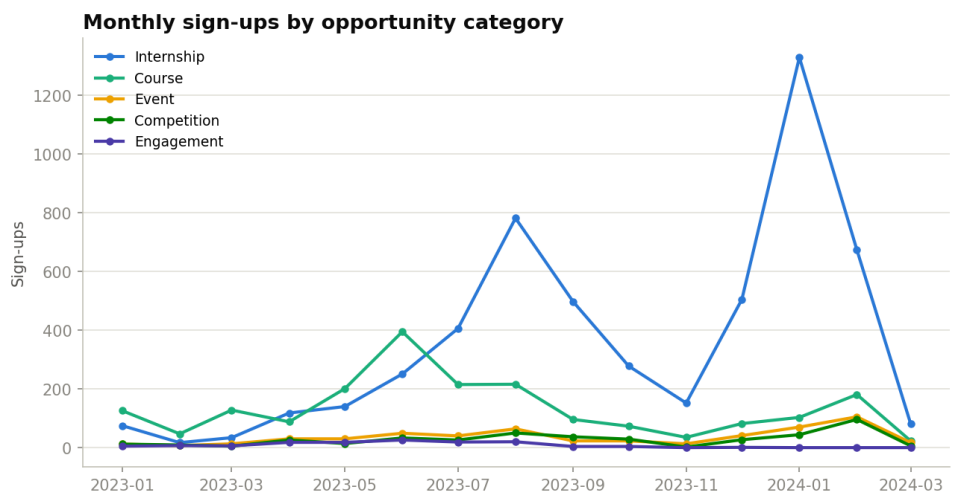


Figure 5 — Monthly sign-ups by opportunity category

Monthly Sign-Up Counts

Month	Sign-ups
Jan 2023	230
Feb 2023	87
Mar 2023	186

Apr 2023	280
May 2023	403
Jun 2023	754
Jul 2023	706
Aug 2023	1,132
Sep 2023	658
Oct 2023	407
Nov 2023	202
Dec 2023	656
Jan 2024	1,547
Feb 2024	1,056
Mar 2024	127

Significant Observations

- **Strong first-year growth:** 230 sign-ups in Jan 2023 to a summer peak of 1,132 in Aug 2023.
- **Autumn slowdown:** monthly declines Sep–Nov 2023, bottoming at 202 (-82% from the August peak).
- **Winter rebound:** Dec 2023 +225%, into the all-time peak of 1,547 in Jan 2024.
- **Upward trajectory:** early-2024 run-rate (~1,300/month) is roughly 2.7x the 2023 average (~475/month).
- **March 2024 is a partial month:** the data ends early March, so its low count is a coverage artifact, not a decline.
- **Internships drive the spikes;** Courses are the steady second contributor.

5. Distribution of Majors Among Applicants

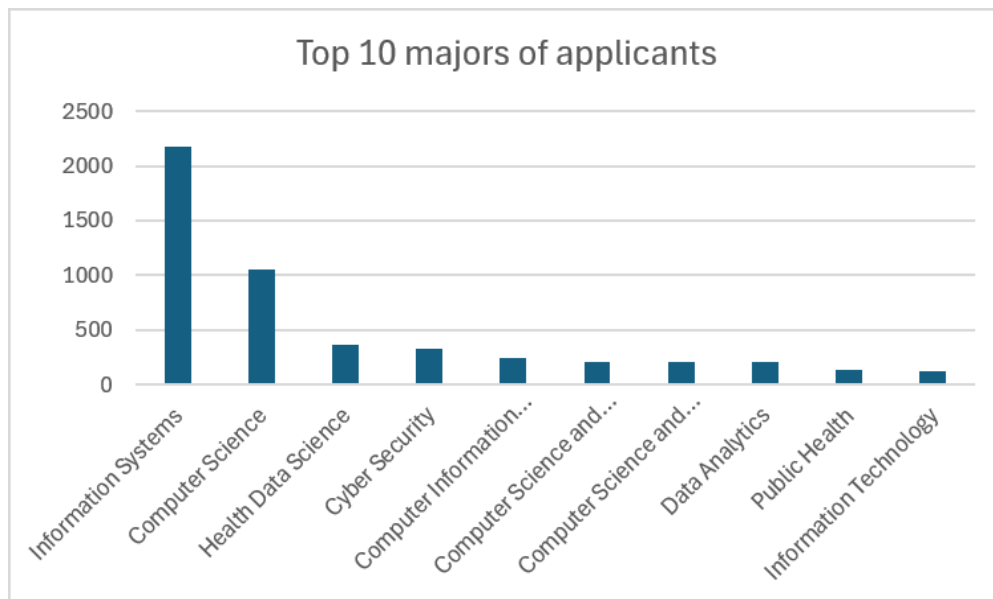


Figure 6 — Top 10 majors of applicants

- **Concentration in tech fields:** Information Systems and Computer Science together account for the large majority of applicants, and several additional top majors are themselves CS/IS hybrid fields (Computer Information Systems, CS and Information Systems, CS and Engineering). This indicates the applicant pool is heavily concentrated in computer- and information-science-adjacent fields rather than being evenly spread across disciplines.

Note: this count was taken across the entire dataset (not broken out per opportunity category) — it reflects the overall applicant cohort's field of study, not category-specific major alignment.

- Given how dominant Information Systems and Computer Science are here, this concentration is likely to reappear in category-specific analysis if that data becomes available.

6. Opportunity Category Analysis

Category Distribution

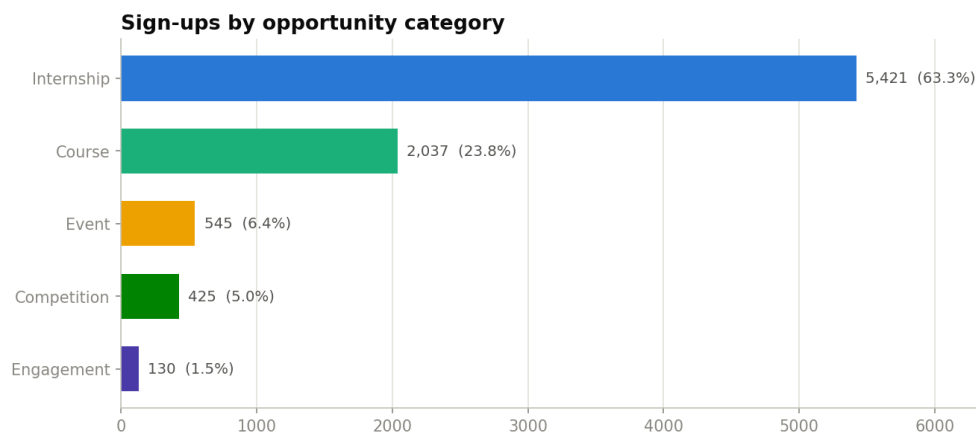


Figure 7 — Sign-ups by opportunity category

Outcome Conversion

Outcomes are aggregated from the raw Status Description field (one status per sign-up record, as recorded at export time). Its eight statuses are grouped by what they mean for the learner:

- Positive = Team Allocated (3,276), Started (767), Rewards Award (29) — passed selection and placed, participating, or completed
- In progress = Applied (105), Waitlisted (109) — no final decision yet
- Negative = Rejected (3,569), Dropped Out (617), Withdraw (86) — did not proceed

A category's positive rate is positive sign-ups divided by all its sign-ups. Statuses are a snapshot, so some in-progress records would eventually resolve either way.

Status-to-Outcome Mapping (Evidence Trail)

Outcome	Status Description	Records
In progress	Waitlisted	109
In progress	Applied	105
Negative	Rejected	3,569
Negative	Dropped Out	617

Negative	Withdraw	86
Positive	Team Allocated	3,276
Positive	Started	767
Positive	Rewards Award	29

Outcome Mix per Category

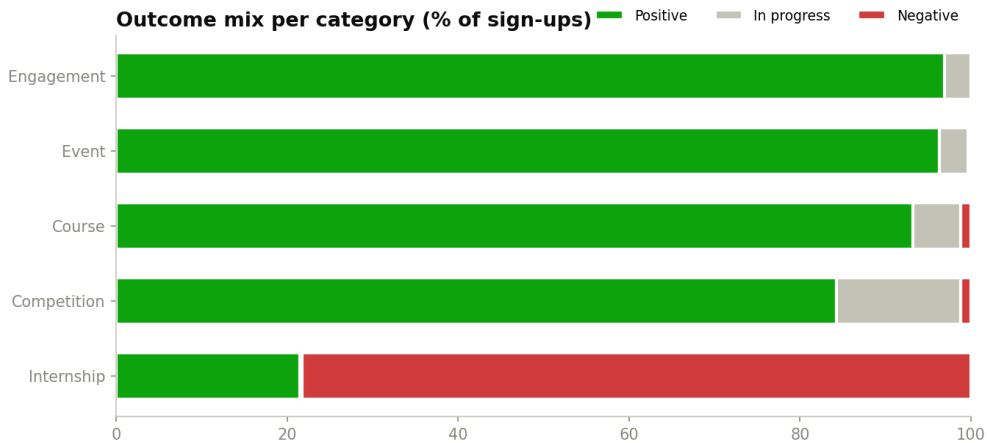


Figure 8 — Outcome mix per category (% of sign-ups)

Category Performance Summary

Category	Sign-ups	Opportunities	Sign-ups/Opp.	Share %	Positive %	Rejection %
Internship	5,421	9	602	63.3	21.5	65.8
Course	2,037	4	509	23.8	93.2	0.0
Event	545	3	182	6.4	96.3	0.0
Competition	425	5	85	5.0	84.2	0.0
Engagement	130	2	65	1.5	96.9	0.0

Significant Observations

- **Internships dominate demand:** 5,421 sign-ups (63.3%), and the highest demand per opportunity (~602 each).
- **Highest conversion:** Engagement (96.9% positive) and Event (96.3%); Courses also strong (93.2%).
- **Lowest conversion: Internship (21.5%).** All 3,569 rejections in the dataset are internship applications (65.8% rejection rate).
- **Lowest volume: Engagement.** 130 sign-ups (1.5% of activity), with only 2 opportunities offered.
- **Implication:** expanding internship capacity or routing rejected applicants to high-conversion Courses/Events are the clearest levers for improving outcomes.

7. Interest Areas for Future Data Collection

Identified as valuable during analysis but not answerable with the current dataset; revisit if additional data becomes available.

- **1. Capacity vs. demand:** How does the volume of applications compare to the total number of available opportunities (capacity) in each category? The dataset records applications and opportunity listings, but not seat capacity or slots available per opportunity.
- **2. Time-to-dropout:** What is the average time elapsed between an opportunity's start date and the point at which accepted students drop out, and how does this vary by category? No timestamp exists for when a dropout occurred, only a status label.
- **3. Release vs. application timing:** How do monthly totals of opportunity releases compare with monthly totals of application volume over the data range? Opportunity

Start Date is missing or invalid for ~54% of records (3,794 missing + 873 invalid of 8,558), making reliable monthly trends infeasible.

- **4. Peak release periods:** Which specific months show the highest concentration of opportunity releases, and how do these peak periods align with application surges? Same limitation as above.
- **5. Supply/demand misalignment:** Where are the most significant misalignments between available opportunities and incoming applications on a monthly basis? Same limitation as above.

Link to Notebook:

https://colab.research.google.com/drive/1MRLy8dGokH6L94fBM9Q1ILKQ2_VI74PY?usp=sharing